# CNC Machinist Course — Master Syllabus

Comprehensive curriculum compiled from top institutions and certification bodies worldwide. Use this as a blueprint to structure your online CNC course.

20 MODULES | 4 LEVELS | FOUNDATION → SPECIALIST

## Source Institutions & Programs

This syllabus synthesises curricula from the following programs, certifications, and industry standards:

- **NIMS (USA)** — National Institute for Metalworking Skills — Level I/II Machining Standards
- **Haas Automation** — HTEC Curriculum & Certification Program
- **MIT OCW** — 2.008 Design & Manufacturing II / 2.810 Manufacturing Processes
- **DMG MORI Academy** — Germany — CNC training for Fachkraft certification
- **Haas + Autodesk Fusion** — CAD/CAM/CNC for 2.5-Axis Milling
- **Thaddeus Stevens College** — Computer Integrated Machining AAS
- **Central Piedmont CC** — Computer-Integrated Machining Technology
- **North Idaho College** — Machining & CNC Technology ATC
- **Surry Community College** — Computer-Integrated Machining — CNC Track
- **Mesa Community College** — CNC Machining I & II Certificates
- **Laney College** — Machine Technology — including 5-axis & GD&T
- **De Anza College** — DMT — Machining & CNC Course Guide
- **ACE — America's Cutting Edge** — DoD-funded CNC & Metrology Training
- **India DGT** — Advanced CNC Machining Technician — NSQF Level 4
- **Workshops For Warriors** — CNC Machining I & II — Veteran Training
- **GCU (Grand Canyon Univ.)** — CNC Machinist Certificate

Level 1 Foundation Modules 01-05 → Level 2 Core CNC Modules 06-11 → Level 3 Advanced Modules 12-17 → Level 4 Specialist Modules 18-20

## LEVEL 1 Foundation

### 01 Shop Safety & Industrial Practices — NIMS, OSHA

Prerequisite for all other modules — covers OSHA, PPE, hazard communication

**SAFETY FUNDAMENTALS**

- → Introduction to OSHA standards for machine shops
- → Personal protective equipment (PPE): eye, ear, hand, foot protection
- → Hazard communication: Safety Data Sheets (SDS), labeling
- → Fire prevention and extinguisher types
- → Lockout/Tagout (LOTO) procedures
- → Electrical safety around machine tools

**MACHINE SHOP PRACTICES**

- → Housekeeping: chip removal, floor cleanliness, coolant management
- → Ergonomics: lifting, posture at machines, repetitive strain
- → Material handling: overhead cranes, forklifts, pinch/crush points
- → Noise reduction and hearing conservation
- → Compressed air safety
- → Emergency procedures and first aid basics

### 02 Shop Mathematics

Fractions, decimals, tolerances, trigonometry for machining

### 03 Blueprint Reading & Technical Drawing

Orthographic projections, dimensioning, tolerancing fundamentals

### 04 Precision Measurement & Metrology — NIMS

Measuring instruments, calibration, inspection techniques

### 05 Materials Science for Machinists

Metals, alloys, heat treatment, material properties

## LEVEL 2 Core CNC Skills

### 06 Manual Machining Fundamentals

Engine lathe, vertical mill, drill press, surface grinder, bench work

### 07 Cutting Tools & Toolholders — NIMS

Tool geometry, insert selection, speeds & feeds, cutting fluids

### 08 Workholding Principles & Fixturing

Clamping, locating, supporting — for both turning and milling

### 09 CNC Machine Operation & Safety — NIMS, Haas

Machine components, controls, startup, basic operation of CNC mills & lathes

### 10 G-Code Programming — Milling — NIMS

Manual G-code for CNC milling centers: 2D and 2.5D operations

### 11 G-Code Programming — Turning — NIMS

Manual G-code for CNC turning centers: OD, ID, threading, grooving

## LEVEL 3 Advanced CNC

### 12 CNC Setup, Optimization & Troubleshooting

Machine setup procedures, cycle time optimization, problem diagnosis

### 13 CAD for Manufacturing — Autodesk

3D modeling with Fusion, SolidWorks, or NX — DFM principles

### 14 Geometric Dimensioning & Tolerancing (GD&T) — ASME Y14.5

Datum systems, form/orientation/location/runout tolerances

### 15 CAM Programming — 2.5 & 3-Axis — Autodesk, Mastercam

Toolpath generation, simulation, post-processing — Fusion / Mastercam / NX

### 16 CAM Programming — Turning & Mill-Turn

Lathe CAM, live tooling, C-axis, Y-axis mill-turn operations

### 17 Quality Systems & Process Control — ISO 9001

Quality management, SPC, documentation, continuous improvement

## LEVEL 4 Specialist Tracks

### 18 Multi-Axis CNC Machining (4 & 5-Axis)

4th axis rotary, 3+2 positional, simultaneous 5-axis strategies

### 19 Advanced Programming & Automation

Macro B / parametric programming, probing, bar feeders, pallet systems

### 20 Industry 4.0 & Emerging Technologies

Digital twins, additive manufacturing, IoT monitoring, career development

## Certification & Reference Alignment

| Certification / Program | Modules Covered | Notes |
|---|---|---|
| **NIMS Level I — CNC Milling Operations** | 01, 02, 03, 04, 07, 08, 09, 10 | Core operator credential — widely recognized in US |
| **NIMS Level I — CNC Turning Operations** | 01, 02, 03, 04, 07, 08, 09, 11 | Paired with milling for full NIMS Level I |
| **NIMS — CNC Mill/Lathe Programming, Setup & Operations** | 01–12 | Requires setup sheet competency + practical exam |
| **Haas CNC Certification** | 01, 09, 10, 11 | Free online + hands-on test at Haas Factory Outlet |
| **Autodesk Certified Associate — CAM for 2.5 Axis Milling** | 03, 13, 15 | Validates Fusion CAD/CAM competency |
| **DMG MORI Academy — Fachkraft** | 09–12, 18 | German-standard industrial training — Siemens controllers |
| **ISO 9001 / AS9100 Awareness** | 04, 14, 17 | Quality systems relevant to aerospace & medical manufacturing |
| **Community College AAS Degree (typical)** | 01–17 | 2-year associate degree covering Levels 1–3 |

## Recommended Textbooks & Free Resources

| Resource | Coverage | Cost |
|---|---|---|
| Kalpakjian — *Manufacturing Processes for Engineering Materials* | Materials, cutting theory, process physics | Textbook |
| Haas Operator's Manual — Programming Chapter | G/M codes, canned cycles, programming examples | Free (haascnc.com) |
| Haas CNC Certification Program | Basic mill & lathe operation, safety | Free online |
| MIT OCW 2.008 / 2.810 | Manufacturing process physics, quality, systems | Free |
| Autodesk Fusion — Self-Paced Learning | CAD modeling, CAM toolpaths, simulation | Free (education) |
| Haas + Autodesk Fusion Curriculum | End-to-end blueprint → CAD → CAM → CNC workflow | Free |
| America's Cutting Edge (ACE) | CNC machining, metrology, machining dynamics | Free online |
| Tooling U-SME (toolingu.com) | NIMS-aligned self-paced courses — all modules | Subscription |
| CamInstructor | Haas CNC training, Mastercam, project-based | Per-student license |